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REPO: <https://github.com/biswajit134/GitOps-School-Management-System.git>

10. GitOps Workflow using ArgoCD on Kubernetes

Introduction:

The School Management System (SMS) is a comprehensive, multi-role web application designed to digitize administrative and academic workflows. While the core application focuses on managing students, faculty, and performance tracking using the MERN stack, the primary objective of this project is to demonstrate modern DevOps maturity. By leveraging Infrastructure as Code (IaC) and GitOps principles, the project ensures that the application is scalable, self-healing, and deployed through a fully automated pipeline into the Azure Kubernetes Service (AKS).

Abstract:

Traditional deployment methods often suffer from configuration drift and manual errors. This project implements a **GitOps workflow** where the "Source of Truth" resides entirely within Git repositories.

- **Application Side:** A React-based frontend and Node.js backend are containerized and analyzed for security and quality.
- **Infrastructure Side:** Azure resources are provisioned via **Terraform**.
- **Delivery Side:** **ArgoCD** acts as the GitOps controller, synchronizing Kubernetes manifests from Git to the AKS cluster. This ensures that any manual changes to the cluster are automatically reverted to the state defined in the repository, providing a robust, auditable, and automated deployment lifecycle.

Tools Used:

Category	Tools Used
IaC	Terraform
Terraform State Management	HashiCorp Cloud Platform
CI/CD Automation	GitHub Actions
Kubernetes Package Manager	Helm
GitOps Controller	ArgoCD
Containerization	Docker
Orchestration	Kubernetes (K8s)
Managed Kubernetes Providers	Azure Kubernetes Service (AKS)
Cloud Provider	Azure

Static Analysis	SonarQube
Security Scanning	Trivy
Artifact Repo	Docker Hub

Steps Involved in Building the Project:

Phase 1: Application Development & Containerization

1. MERN Development: Developed the three-tier architecture (Admin, Teacher, Student modules).
2. Database Integration: Configured MongoDB Atlas for a managed, scalable database layer.
3. Dockerization: Authored optimized Dockerfiles for both Frontend and Backend. Used multi-stage builds to reduce image size and minimize the attack surface by excluding build-time dependencies from the final production image.

Phase 2: Infrastructure Provisioning

1. Terraform Configuration: Wrote HCL scripts to define the Azure Resource Group, Virtual Network, and AKS Cluster.
2. State Management: Integrated Terraform Cloud to maintain the terraform.tfstate file, enabling team collaboration and versioned infrastructure.

Phase 3: Continuous Integration (CI)

1. Workflow Automation: Created GitHub Actions to trigger on code pushes.
2. Code Analysis: Implemented syntax checking and SonarQube scans to ensure code maintainability.
3. Artifact Management: Automated the building and pushing of Docker images to Docker Hub with unique tags to facilitate traceable deployments.

Phase 4: Continuous Delivery & GitOps (CD)

1. Manifest Preparation: Defined Kubernetes Deployments, Services, and ConfigMaps in a dedicated manifest directory.
2. ArgoCD Setup: Deployed ArgoCD within the AKS cluster and linked it to the GitHub repository.
3. Automated Sync: Configured the "Pull" model where ArgoCD detects manifest updates in Git and automatically rolls out changes to the cluster.

Conclusion:

The implementation of the School Management System via a GitOps pipeline successfully demonstrates the power of automation in modern software engineering. By treating Infrastructure as Code and utilizing a GitOps controller (ArgoCD), the project achieves high availability, environment consistency, and a simplified disaster recovery process. This architecture reduces the "time-to-market" for new features while maintaining strict security and quality gates through the CI/CD pipeline.